

Technical Assessment: Sentinel-2 Change Analysis

Context

At our company, we build geospatial analytics products using satellite imagery, especially optical data such as Sentinel-2.

These analytics are used by governments and enterprise customers to monitor land-surface changes, detect anomalies, and support operational decision-making.

This assessment evaluates your ability to:

- Work with real satellite data
- Design geospatial data pipelines
- Implement change detection
- Store and query geospatial analytics
- Visualize and interpret results

We care more about your reasoning, clarity, and engineering quality than scientific accuracy.

Provided Data

You receive the following files:

```
inputs/  
  aoi.geojson           # Area of interest  
  example_change_detection.py # Example change detection algorithm  
data/  
  sentinel2_20230812/    # Sentinel 2 data folder (3 bands), date 1  
  sentinel2_20230902/    # Sentinel 2 data folder, date 2
```

Each Sentinel-2 folder contains 3 bands corresponding to optical imagery:

- Band 2 (Blue)
- Band 3 (Green)
- Band 4 (Red)

These bands are already clipped to the area of interest (AOI). And the AOI is located at an open-pit mining site in Zambia.

High-Level Objective

Build a small geospatial analytics pipeline that:

1. Loads Sentinel-2 imagery from the provided folders
2. Performs change detection between the two dates
3. Stores detected changes in a geospatial database
4. Visualizes the detected changes
5. Interprets and communicates the results

PART 1: Data Preparation

Goal: Evaluate your ability to correctly load and prepare multi-band satellite imagery.

Task

Using the provided Sentinel-2 folders:

Load Bands 2, 3, and 4 for both dates.

You should:

- Read the raster data
- Verify CRS, transform, and dimensions consistency
- Stack the three bands into a multi-band image for each date

Example output structure (in memory or saved):

```
data/processed/  
sentinel2_20230812_stack.tif  
sentinel2_20230902_stack.tif
```

PART 2: Change detection

Goal

Evaluate geospatial modeling and analytical skills.

Task

Implement a change detection algorithm between the two Sentinel-2 dates using the 3-band imagery.

You may use any method.

Examples include:

- Band difference
- Image ratio
- Vegetation index difference (NDVI if you derive it using other bands optionally)
- Threshold-based detection
- Machine learning or statistical methods

You may use the provided example script as inspiration, but do not simply copy it. Apply the example algorithm, and yours.

Required Output

Produce the .tif output rasters, for example:

```
data/processed/change_map.tif
```

Georeferenced raster representing change intensity.

Also produce the respective binary change .tif maps, for example:

```
data/processed/change_binary.tif
```

Binary raster representing detected change vs no change.

PART 3: Change Feature Extraction and Storage

Goal

Evaluate your ability to convert raster analytics into structured geospatial datasets.

Task

Convert detected change areas into vector features.

Example: Polygons representing areas of detected change.

Store results in database

Use either:

- SQLite (Spatialite optional)

OR

- PostgreSQL/PostGIS

Database Table Example

```
change_features:
```

```
id | date_before | date_after | area_m2 | confidence | geometry
```

Geometry must be stored as geometry type if used.

PART 4: Visualization (Optional, but recommended)

Goal

Evaluate ability to communicate results visually.

Task

Visualize the detected changes.

- You may use:
- Matplotlib
- GeoPandas
- Folium
- Dash Plotly
- Any other visualization tool

Show:

- AOI polygon

- Change raster and/or polygons

Interactive visualization is a plus but not required.

PART 5: Analysis and Interpretation

Goal

Evaluate your ability to interpret geospatial analytics results.

Write a short report

Create: `report.md`

Include the following sections:

1. Method

Explain:

- Change detection algorithm chosen
- Why this method

2. Results

Describe:

- Where changes occur
- Pattern of changes observed

3. Interpretation

Examples:

What might these changes represent?

Examples:

- Urban expansion
- Vegetation change
- Land clearing
- Seasonal variation
- Noise or artifacts

Deliverables

Submit GitHub repo containing:

- Code (Python scripts or notebooks)
- Database (SQLite file OR instructions for PostGIS)
- README.md (explaining how to run the code + your approach + any assumptions)
- `report.md`
- any other files you think is relevant